User Manual for Cloud Sense Vis

Website: https://www.cloudsensevis.com/
Platform: Cloud Sense Vis Environmental & Weather Monitoring

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1. Introduction

Cloud Sense Vis Logo

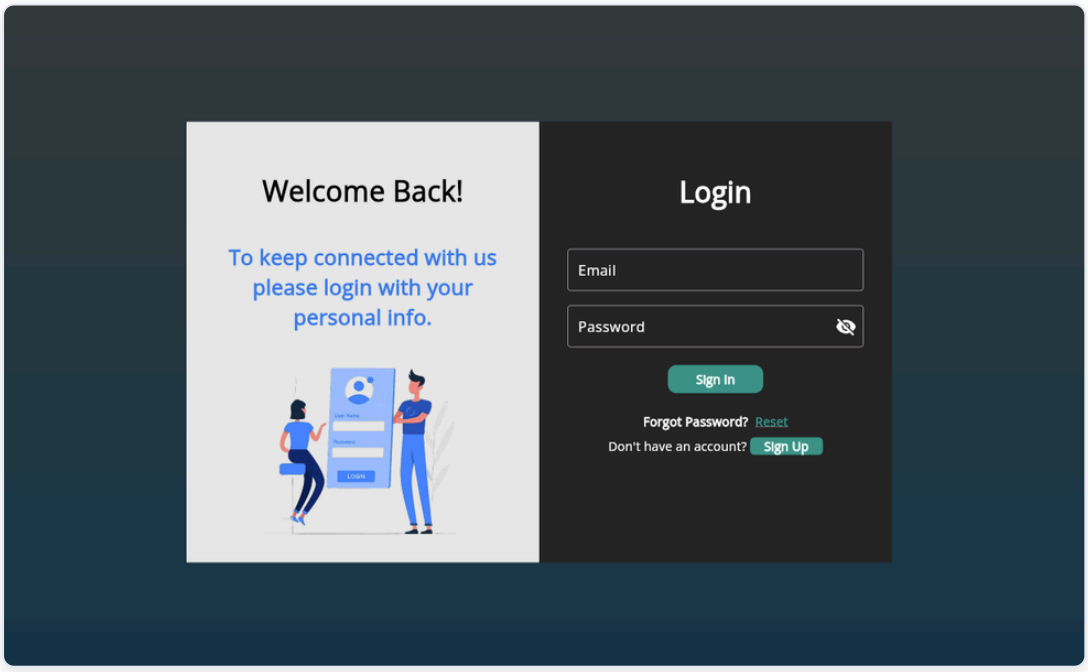
Cloud Sense Vis is an intuitive, web-based monitoring platform designed for real-time tracking of weather and environmental conditions. It connects field-deployed automatic weather stations, smart environmental probes, soil sensors, and livestock collars to an easy-to-use digital dashboard.

With Cloud Sense Vis, users can observe live weather updates, inspect computed environmental parameters on derived weather placards, explore interactive historical trends, pinpoint station locations on a map, download telemetry spreadsheets, and browse compatible hardware solutions all from a single portal.

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2. Login, Registration & Settings

To access Cloud Sense Vis, open your browser and navigate to https://www.cloudsensevis.com/.



Cloud Sense Vis Login Screen

2.1 Logging In

1. Enter your registered **Email** and **Password** into the login card.
2. Click the **Sign In** button to open the main dashboard.

2.2 Registering a New Account

1. If you are a new user, click the **Sign Up** link at the bottom of the login box.
2. Enter your email address and create a secure password.
3. Complete the verification prompt sent to your email to activate your account.

2.3 Password Recovery

1. If you forget your password, click **Forgot Password?** on the login screen.
2. Enter your registered email address and submit.
3. Check your inbox for the password reset instructions to establish a new password.

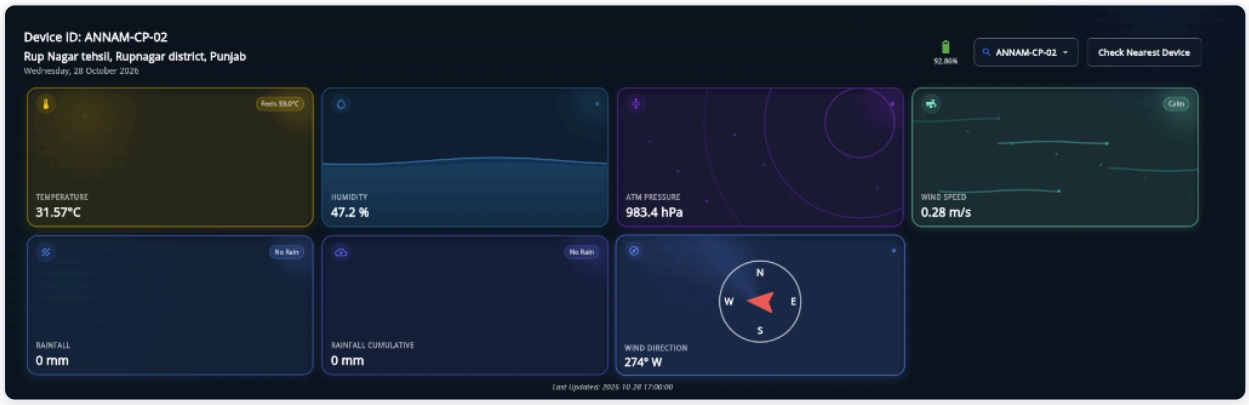
2.4 Theme Customization (Light & Dark Mode)

Located at the top right of every screen, the **Theme** button allows you to switch between: * **Dark Mode:** A dark background designed for low-light environments, field nights, and control centers. * **Light Mode:** A clean, bright display suited for daytime office use.

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3. Main Dashboard Overview

Upon logging in, the **Dashboard** displays live weather telemetry from your active station along with quick station-switching and power indicators.



Cloud Sense Vis Live Dashboard

3.1 Station Header & Controls

- **Device Identification:** Displays the active Device ID (e.g., `ANNAM-CP-02`) and geographical location (e.g., *Rup Nagar tehsil, Rupnagar district, Punjab*) along with the active date.
- **Battery Indicator:** Real-time percentage indicator of field station power (e.g., `92.86%`).
- **Device Selector Dropdown:** Allows immediate switching between different monitoring stations.
- **Check Nearest Device:** Quickly locates and views the closest neighboring monitoring node.

3.2 Real-Time Telemetry Cards

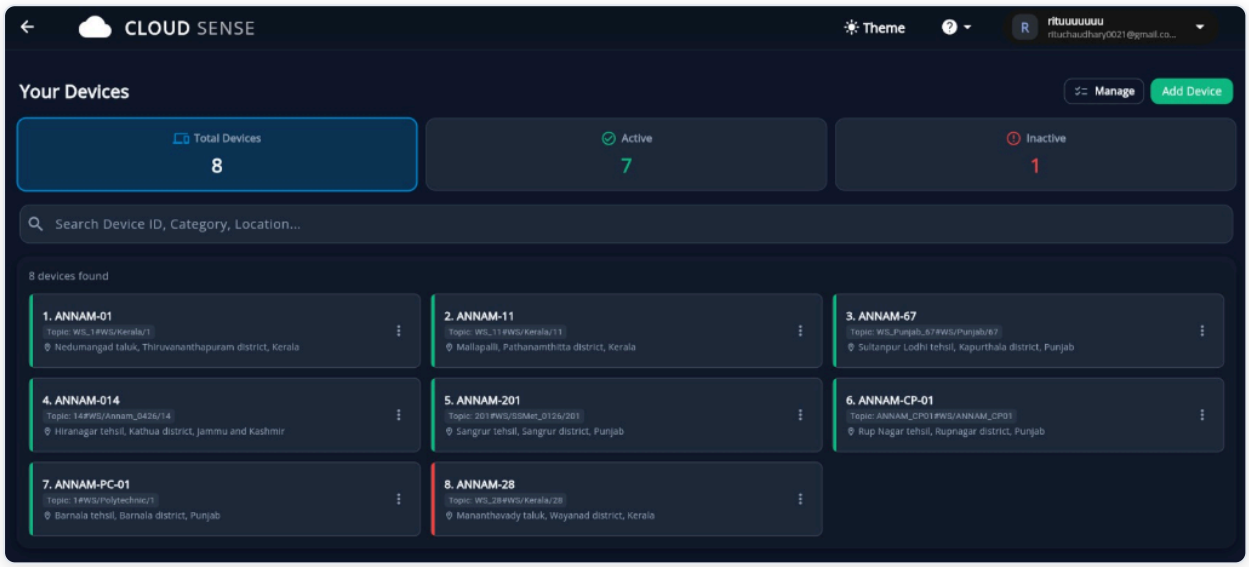
The dashboard presents live environmental measurements across intuitive color-coded cards: * **Temperature:** Current air temperature (e.g., `31.57°C`) with an apparent perceived temperature indicator (`Feels 33.0°C`). * **Humidity:** Real-time relative moisture percentage in the air (e.g., `47.2 %`). * **Atmospheric Pressure:** Barometric pressure calibrated to local elevation (e.g., `983.4 hPa`). * **Wind Speed:** Instantaneous wind velocity (e.g., `0.28 m/s`) with atmospheric state classification (`Calm`). * **Rainfall:** Current incremental rainfall reading (e.g., `0 mm`, with a `No Rain` badge). * **Rainfall**

Cumulative: Total accumulated precipitation over the recording cycle (e.g., `0 mm`). * **Wind Direction:** Displays the dynamic compass heading angle and cardinal orientation (e.g., `274° W`) with an integrated directional compass dial. * **Last Updated:** Timestamp at the bottom indicating the exact time the last data packet was ingested.

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4. Managing Your Devices

Click on **Your Devices** to access the complete fleet management console showing all registered hardware units.



Your Devices Fleet Overview

4.1 Fleet Summary Counters

At the top of the screen, three overview cards summarize your network health at a glance: * **Total Devices:** Total count of hardware units provisioned under your account (e.g., 8). * **Active:** Number of devices currently online and transmitting data normally (e.g., 7, highlighted in green). * **Inactive:** Number of devices that are currently offline or delayed in reporting (e.g., 1, highlighted in red).

4.2 Search and Action Controls

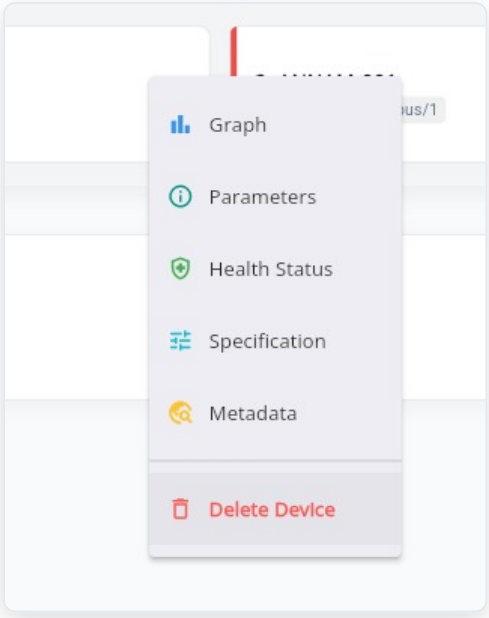
- **Search Bar:** Instantly filter devices by typing a Device ID, hardware category, or geographical location (e.g., *Search Device ID, Category, Location...*).
- **Manage Button:** Opens fleet administration tools for bulk settings and configuration.
- **Add Device Button:** Green shortcut button at the top right to register a new monitoring node.

4.3 Device Grid & Card Layout

Each registered station appears as an organized card displaying: * **Index & Device ID:** Identifier (e.g., `1. ANNAM-01`, `2. ANNAM-11`, `3. ANNAM-67`, `6. ANNAM-CP-01`). * **Topic Channel:** MQTT transmission topic for data routing (e.g., `WS_1#WS/Kerala/1`, `WS_Punjab_67#WS/Punjab/67`). * **Location Tag:** Exact taluk, district, and state where the device is installed (e.g., *Nedumangad taluk, Thiruvananthapuram district, Kerala* or *Sangrur tehsil, Sangrur district, Punjab*). * **Status Stripe:** A vertical color stripe on the left edge of each card provides instant status visibility: * **Green Stripe:** Device is **Active** and operating normally. * **Red Stripe:** Device is **Inactive** and requires maintenance or network checking.

4.4 Device Actions & Options (Three-Dot Context Menu)

Clicking the three vertical dots on the right side of any device card opens a comprehensive device management menu:



Device Actions Context Menu

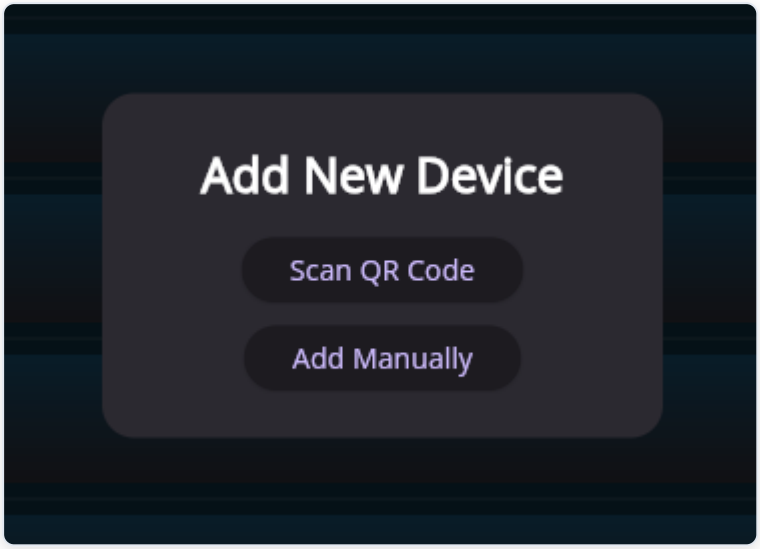
This menu provides immediate access to six vital operational functions:

1. **Graph:**
2. Directly opens the dedicated **Device Details & Graphs** page for this station.
3. Allows you to inspect historical curves, view daily min/max summaries, and export data in CSV format.
4. **Parameters:**
5. Displays all active multi-channel sensor readings for this node in a clean list view.
6. Shows current numerical values, engineering units, and channel timestamps across all connected physical probes.
7. **Health Status:**
8. Opens the **Device Health Status** diagnostic panel.
9. Displays battery voltage, solar power charging levels, cellular transmission regularity, and link status to help you diagnose hardware or network issues proactively.
10. **Specification:**
11. Fetches the complete technical specifications for the selected unit.
12. Details the hardware model, installed sensors (e.g., Temperature/Humidity sensor type, Wind Sensor model, Rain Gauge type), operating ranges, and institutional compliance standards.
13. **Metadata:**
14. Retrieves registered backend metadata associated with the station.
15. Displays the MQTT transmission topic, geographic latitude and longitude coordinates, administrative taluk/district tags, and device registration timestamps.
16. **Delete Device:**
17. Allows you to safely unbind and remove the device from your account.
18. Opens a confirmation prompt to prevent accidental deletions while keeping historical database archives intact.

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5. Adding a New Device

To register a new station or sensor to your account, click the green **Add Device** button at the top right of the device screen.



Add New Device Modal

The platform provides two simple methods to connect your hardware:

5.1 Option 1: Scan QR Code

1. Click **Scan QR Code**.
2. Allow camera permissions when prompted by your browser or smartphone.
3. Position your camera over the official QR code label printed on the device or its packaging.
4. The system decodes the hardware key and automatically associates the device with your account.

5.2 Option 2: Add Manually

1. Click **Add Manually**.
2. Type in your assigned Device ID (e.g., `CP001`, `WD101`, `SS005`, or `CL102`).
3. Click **Add Device** to finalize. A confirmation prompt confirms successful association.

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6. Interactive Graphs & Historical Data Analysis

Selecting any device opens its dedicated **Device Details** page, providing real-time operational status, parameter summary statistics, and interactive historical trend graphs.



Device Details Interactive Graphs

6.1 Station Header & Live Connectivity

- **Hardware Info:** Displays the Device ID (e.g., `ANNAM-01`), firmware/software version (`v1.0.0`), and geographic location.
- **Status Badges:** Shows last active date and time (e.g., `28-10-2026 17:00:00`), live status pill (`● Live`), cellular signal strength bars, solar charging status, battery health icon, and a manual data refresh button.

6.2 Parameter Summary Cards

Top parameter cards provide instantaneous readings alongside minimum and maximum historical bounds for the day: * **Temperature:** Current temperature (e.g., `32.13 °C`) with recorded minimum (`▼ 25.89`) and maximum (`▲ 39.67`). * **Humidity:** Relative humidity (e.g., `65.20 %`) with diurnal bounds (`▼ 44.87`, `▲ 88.31`). * **Rainfall:** Incremental rainfall (e.g., `0.00 mm`) and total accumulated rain (`0.00 mm`). * **Wind:** Current wind speed (e.g., `0.56 m/s`) with directional heading (`145° South East`). * **Max Wind Gust:** Peak recorded gust velocity (e.g., `3.01 m/s`) with exact timestamp of occurrence. * **Atmospheric Pressure:** Calibrated barometric pressure (e.g., `1000.09 hPa`) with min/max tracking (`▼ 1000.01`, `▲ 1004.01`).

6.3 Timeframe Selectors & Display Controls

- **Time Range Buttons:** Easily toggle between **Date Picker** (`28/10/2026`), **7 Days**, **1 Month**, **3 Months**, and **1 Year**.
- **Graphs per Row:** Adjust chart density to view single or multiple graph columns.
- **Compare Sensors:** Compare measurements across multiple adjacent sensors on a unified timeline.

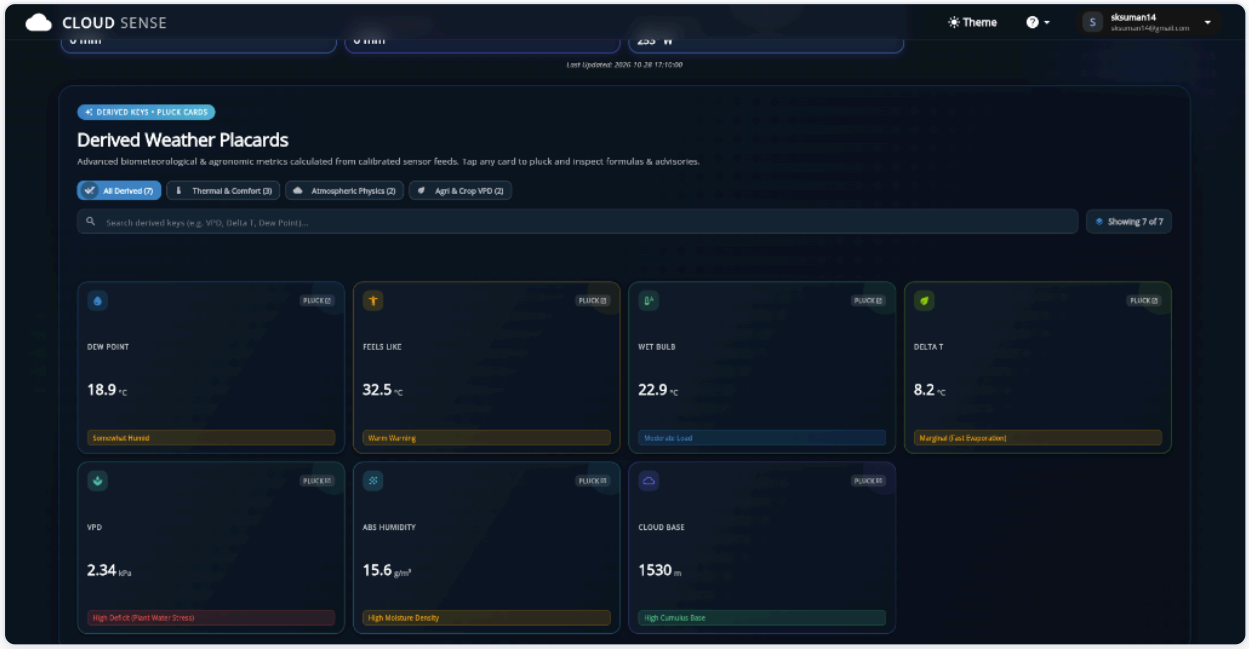
6.4 Interactive Trend Charts & Data Export

- **Time-Series Curves:** Interactive graphical curves (such as Temperature and Humidity) plotting measurements against time intervals.
- **Individual Chart Download:** Click the download icon on the top right of each chart title to export the individual visual graph.
- **One-Click Telemetry Export:** Click the circular floating download button in the bottom right corner to export the entire dataset in `.csv` spreadsheet format for reporting and analysis.

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7. Derived Weather Placards

Cloud Sense Vis includes an advanced biometeorological computational engine called **Derived Weather Placards**. This module calculates secondary atmospheric, agronomic, and human comfort metrics from raw sensor feeds.



Derived Weather Placards

7.1 Placard Categories & Search

- **Category Filter Pills:** Quickly filter cards by functional domain:
- **All Derived (7):** Displays all calculated placard cards.
- **Thermal & Comfort (3):** Human and animal heat stress indices (*Dew Point, Feels Like, Wet Bulb*).
- **Atmospheric Physics (2):** Physical air mass and ceiling metrics (*Abs Humidity, Cloud Base*).
- **Agri & Crop VPD (2):** Crop transpiration and agricultural spraying indicators (*VPD, Delta T*).
- **Search Bar:** Search derived parameters dynamically by keyword (e.g., *Search derived keys (e.g. VPD, Delta T, Dew Point)...*).

7.2 Detailed Breakdown of the 7 Derived Placards

1. **Dew Point (`18.9 °C`)**
2. **Status Tag:** `Somewhat Humid`
3. **Meaning:** The temperature at which air reaches 100% relative humidity and moisture condenses into dew, fog, or frost.
4. **Application:** Used by agriculturalists to forecast morning frost risk and prevent condensation buildup inside greenhouses.
5. **Feels Like (`32.5 °C`)**
6. **Status Tag:** `Warm Warning`
7. **Meaning:** The apparent temperature felt on exposed skin when humidity is factored into ambient temperature.
8. **Application:** Protects outdoor field workers and livestock from heat-related illnesses and guides work-rest cycles.
9. **Wet Bulb (`22.9 °C`)**
10. **Status Tag:** `Moderate Load`

11. **Meaning:** The lowest temperature achievable purely through the evaporative cooling of water at the current atmospheric state.
12. **Application:** Critical for calculating the Wet Bulb Globe Temperature (WBGT) index and sizing evaporative cooling systems in barns and storage sheds.
13. **Delta T (`8.2 °C`)**
14. **Status Tag:** `Marginal (Fast Evaporation)`
15. **Meaning:** The difference between dry air temperature and wet bulb temperature ($\Delta T = T - T_{wb}$).
16. **Application:** The global standard for pesticide and fertilizer spray operations:
    - **Ideal Window ($2^\circ\text{C to }8^\circ\text{C}$):** Optimal droplet survival and coverage.
    - **Marginal / High ($> 8^\circ\text{C}$):** Spray droplets evaporate too quickly in mid-air, causing drift and reduced efficiency.
17. **Vapor Pressure Deficit — VPD (`2.34 kPa`)**
18. **Status Tag:** `High Deficit (Plant Water Stress)`
19. **Meaning:** The difference between the pressure of water vapor inside a leaf and in the surrounding air.
20. **Application:** Governs plant transpiration and nutrient absorption:
    - **Optimal ($0.8\text{ to }1.2\text{ kPa}$):** Healthy growth.
    - **High Deficit ($> 1.6\text{ kPa}$):** Air is dry; crops close their stomata to prevent dehydration, leading to water stress.
21. **Absolute Humidity (`15.6 g/m³`)**
22. **Status Tag:** `High Moisture Density`
23. **Meaning:** The actual mass of water vapor present in one cubic meter of air, independent of temperature.
24. **Application:** Essential for crop drying, grain storage ventilation, and industrial HVAC dehumidification.
25. **Cloud Base (`1530 m`)**
26. **Status Tag:** `High Cumulus Base`
27. **Meaning:** The estimated altitude above ground level where convective clouds start forming (Lifting Condensation Level).
28. **Application:** Valuable for aviation, micro-meteorology, and local weather forecasting.

7.3 The "PLUCK ↗" Feature

Every placard card includes a dedicated **PLUCK ↗** button in its top right corner. Clicking **PLUCK** expands the card, allowing users to inspect the underlying mathematical formula, calibrated thresholds, and agronomic advisories associated with that specific metric.

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8. Map & Location View

The platform includes a built-in map view to help you see the geographic distribution of your stations.



Interactive Map and Deployment View

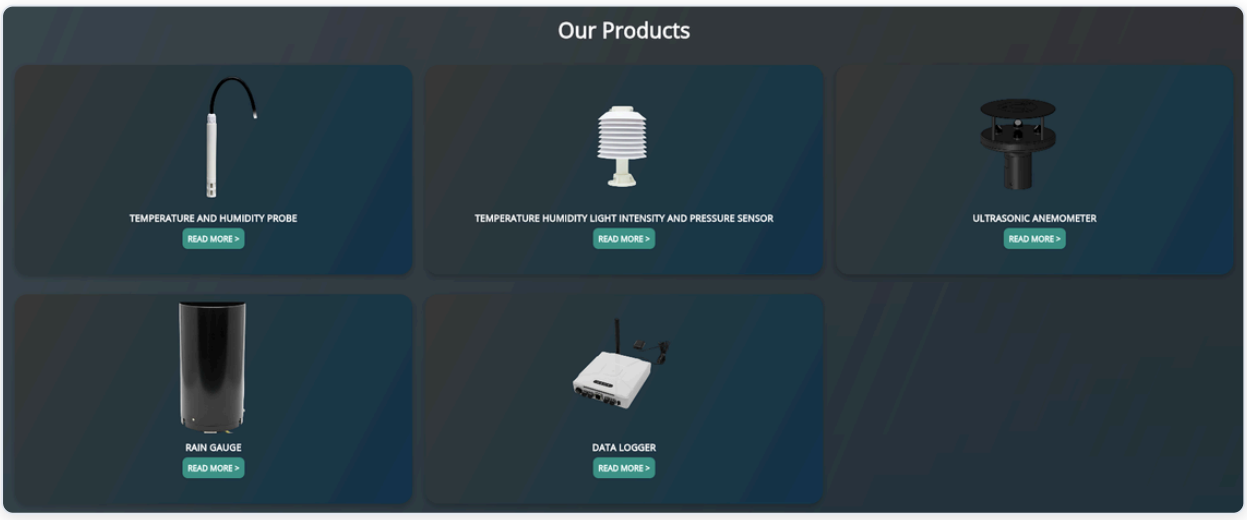
8.1 Map Capabilities

- **Interactive Station Pins:** See all your stations placed on the map with status-coded pins (green for online, red for offline).
- **Regional Filters:** Filter your view across states, districts, taluks, and local village panchayats to analyze regional weather differences.
- **Spatial Data Export:** Download location-aware weather datasets containing geographic coordinates.

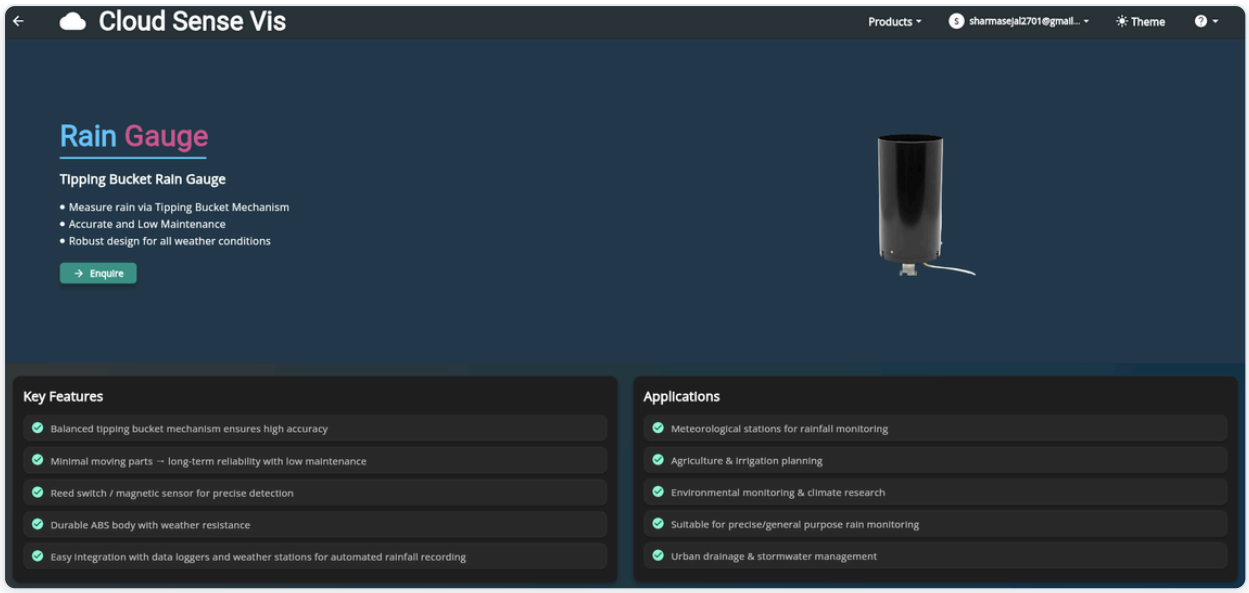
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9. Products & Hardware Range

All hardware products compatible with Cloud Sense Vis are conveniently presented in a single, unified **Products** section accessible from the top navigation bar.



Our Products Catalog Grid



Product Detail Page with Enquire Button

9.1 How to Explore & Enquire

- **Browsing Products:** The catalog displays all available sensing hardware with clear photos, product names, and primary categories.
- **Viewing Product Details:** Click **Read More** on any product card to open its dedicated information page, which explains how the device works, its key advantages, and common practical applications.
- **Submitting an Inquiry:** Every product page features a prominent **Enquire** button. Clicking this button opens a direct contact form allowing you to request quotes, technical guidance, or procurement assistance from the team.

9.2 Summary of Available Hardware Solutions

Product	Primary Purpose	How It Works & Where It Is Used
Rain Gauge	Precipitation Measurement	Uses an internal balanced tipping bucket mechanism that tips with rainfall. Ideal for farms, flood warning systems, and city drainage.
Data Logger	Central Brain & Cloud Uplink	Collects readings from all connected sensors, saves them locally, and transmits them to the cloud over dual-SIM 4G mobile networks.
Ultrasonic Anemometer	Wind Speed & Direction	Measures wind velocity and direction using ultrasonic sound pulses with zero moving parts. Ideal for weather stations and wind farms.
Temperature, Humidity, Light & Pressure Sensor	Multi-Parameter Weather Station	Combines 4 key weather sensors inside a louvered white radiation shield that blocks solar heat while ensuring natural airflow.
Temperature & Humidity Probe	Extended Ambient Probe	Heavy-duty probe on a flexible cable, protected by a sintered filter tip for greenhouses, ducts, and crop canopies.
BLE Wireless Gateway	Wireless Sensor Aggregator	Connects up to 100+ low-power wireless sensor tags over long-range Bluetooth (up to 1 km) and uploads data to the cloud.
7-in-1 Soil Sensor	Subsurface Soil Intelligence	Waterproof probe with stainless steel pins that measures soil moisture, temperature, salinity (EC), pH, and N-P-K nutrients.
Water Quality Monitor (Chloritron)	Water Cleanliness & Safety	Submersible probe monitoring residual chlorine, pH, turbidity, and cleanliness for drinking water networks and aquaculture.
Livestock Smart Tags	Animal Activity & Health	Smart animal collars that track rumination time, movement activity, and temperature for cattle and buffalo herd health.

10. General Installation & Field Setup Guidelines

Proper on-site installation ensures your weather station operates reliably and captures accurate data.



Field Weather Station Installation

10.1 Site Selection

- Choose an open, level location clear of tall trees, walls, or roofs that could obstruct wind or cast shadows on the solar panel.
- Ensure the mounting pole is firmly anchored with concrete or strong ground clamps so it remains completely steady during storms.

10.2 Sensor Mounting Order

- **Wind Sensor:** Always mount the Ultrasonic Anemometer at the **highest point of the mast** to measure uninterrupted wind flow.
- **Rain Gauge:** Mount lower on the stand, ensuring the top funnel is perfectly level and clear of any overhead drip hazards.
- **Solar Panel:** Face the solar panel towards the sun (facing South in India) and tilted to catch maximum daily sunlight.

10.3 Crucial Pre-Check: Rain Gauge Seesaw Lock

- When unpacking the Rain Gauge, open the outer cylinder and **ensure the transit rubber packing band around the internal seesaw bucket is removed**. The bucket must be able to swing freely back and forth to count rainfall correctly.

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11. Standards, Quality & Support

Cloud Sense Vis is built in adherence to recognized scientific and quality benchmarks.

| IMD Weather Guidelines | NABL Accredited Testing | | :---: | :---: | |

IMD Logo

IMD Logo

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NABL Logo

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- **Development:** Developed by **ANNAM.AI** at the Technology Business Incubator (TBI), Indian Institute of Technology (IIT) Ropar, Punjab, India.
- **Meteorological Alignment:** Designed in compliance with India Meteorological Department (IMD) observation norms.
- **Certified Calibration:** Tested and calibrated in NABL-accredited environmental test chambers.
- **Contact & Assistance:**
- **Website:** https://www.cloudsensevis.com/
- **Product Inquiries:** Click the **Enquire** button on any product page.
- **Location:** ANNAM.AI, TBI, IIT Ropar, Rupnagar, Punjab, India.